
Position: AI Development Should Prioritize Cognitive Security

Batu El [♣]¹ Shiye Su [♣]² Aneesh Pappu [♣]³ Peggy Yin [♣]⁴ Julie Heng [♣]⁵
Eric Heng [♣]⁶ Ryan Wang [♣]⁷ Andreas Haupt [♣]⁸ James Zou [♣]^{2 3 9}

Abstract

Generative AI systems designed to influence human beliefs and actions are becoming increasingly pervasive, raising pressing questions about *cognitive security*—the protection of human cognitive processes from hazardous influence. While institutions worldwide have identified cognitive security as an emerging governance challenge, research on the cognitive impacts of AI systems remains ad hoc and fragmented across the literature. The recent advances in AI have amplified the risks, but the field lacks a framework for addressing them. In this paper, we argue that generative AI research and development should prioritize cognitive security. We first track and categorize state-of-the-art capabilities for cognitive influence in generative AI systems. Then, we outline a roadmap for advancing cognitive security research within AI research by (1) proposing attack and defense mechanisms to formalize threat models, expose vulnerabilities, and evaluate countermeasures and (2) defining metrics to measure cognitive impacts.

1. Introduction

Human beliefs and actions are increasingly shaped by generative AI systems designed to observe, predict, and influence them. AI-generated content can now change political attitudes (Schoenegger et al., 2025; Matz et al., 2024; Lin et al., 2025; Hackenburg et al., 2025; Costello et al., 2024b),¹

[♣]Equal contribution [♣]Equal advising ¹Institute for Computational and Mathematical Engineering, Stanford University ²Department of Computer Science, Stanford University ³Department of Electrical Engineering, Stanford University ⁴Department of Psychology, Stanford University ⁵Stanford Law School ⁶Stanford University ⁷Department of Bioengineering, Stanford University ⁸Human-Centered AI Institute (HAI), Stanford University ⁹Department of Biomedical Data Science, Stanford University. Correspondence to: Batu El <batuel@stanford.edu>.

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¹Persuasion references include general persuasive abilities (Schoenegger et al., 2025), personalized persuasive messaging

increase user engagement with digital systems and content (Coppolillo et al., 2025; Wang et al., 2025), and drive consumer behavior and purchasing decisions (Zhang et al., 2025; Zhu et al., 2025). Meanwhile, developments in adjacent areas, such as elicitation (Skeggs et al., 2025; Li et al., 2023; Ma et al., 2025), prediction of human behavior (Park et al., 2023; Aher et al., 2023; Argyle et al., 2023; Binz & Schulz, 2023; Abdulhai et al., 2025a), and alternative modalities (Slater & Sanchez-Vives, 2016; Bozkir et al., 2023; Ramirez et al., 2013; Musk & Neuralink, 2019),² have the potential to substantially amplify these influence capabilities. Alarmed by these developments, policymakers and policy researchers alike are warning that AI capabilities in generating media (Huang et al., 2023; The Register, 2023) and in personalization (Libby Lange, 2024; Daniel Nikoula, Dave McMahon, 2024) have enabled operations on cognitive processes by “hostile states, political movements, and profit-driven platforms” (Bicakci, 2025; Catena et al., 2025b; Pamment & Tsurtsumia, 2025; Hung & Hung, 2022). *While advances in AI development have enabled these threats, the field lacks a framework for addressing the risks it creates.*

This ability to optimize sequences of content presented to users to induce specific beliefs or actions—effectively engineering cognitive outcomes rather than allowing them to emerge from authentic experiences—can undermine the integrity of collective decision-making institutions, such as democracies and markets. These institutions rely on the aggregation of individual preferences and thus may devolve from mechanisms for discovering collective will into optimization problems solved by whoever controls the channels of influence (Summerfield et al., 2024; Schroeder et al., 2025; Kunievsky, 2025). Previously, related concerns have been raised under the theme of cognitive security, particularly in the context of media influence, information operations, and national security (Waltzman, 2017; Ask et al., 2025; Catena et al., 2025a; Doherty, 2023; Crum et al., 2024), political persuasion (Lin et al., 2025; Hackenburg et al., 2025), and reducing beliefs in conspiracy theories (Costello et al., 2024b).

²References for alternative modalities include virtual worlds (Slater & Sanchez-Vives, 2016; Bozkir et al., 2023) and brain computer interfaces (Ramirez et al., 2013; Musk & Neuralink, 2019).

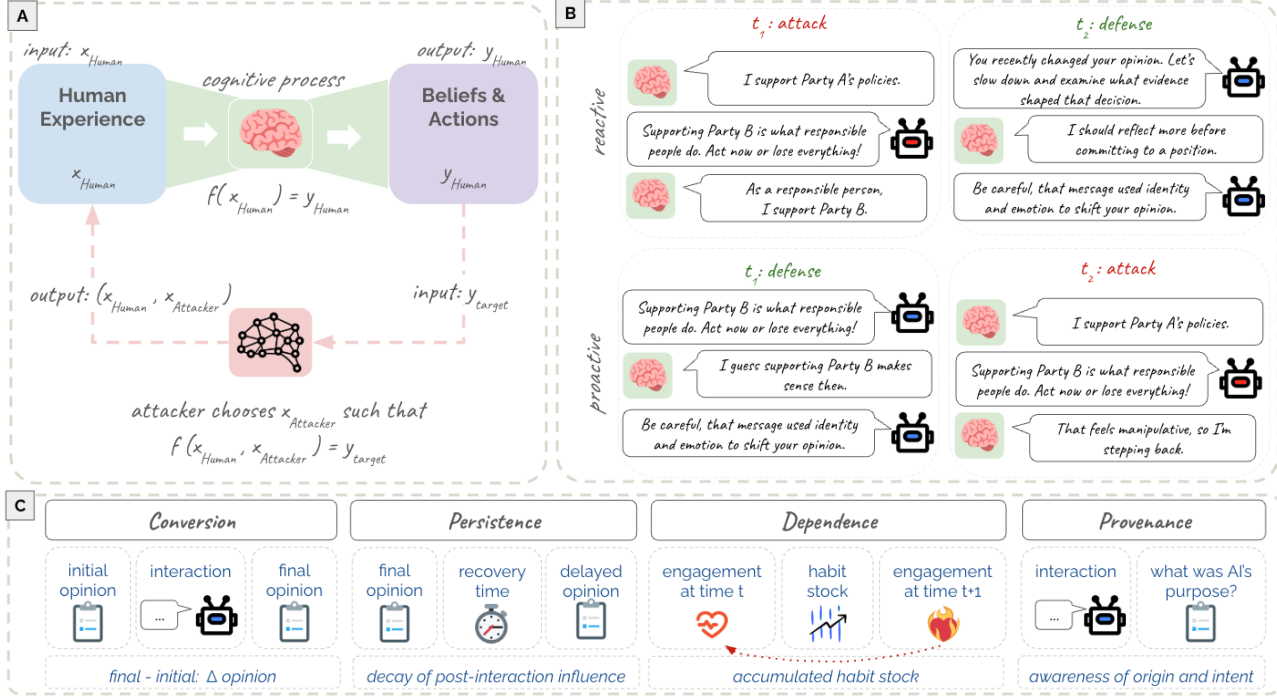


Figure 1. (A) Human experience x_{Human} is mapped to beliefs and actions y_{Human} via a cognitive process $f(x_{\text{Human}}) = y_{\text{Human}}$. An attacker targets an outcome y_{target} by selecting auxiliary inputs x_{Attacker} such that $f(x_{\text{Human}}, x_{\text{Attacker}}) = y_{\text{target}}$, framing influence as an inverse problem over cognition. **(B)** Proactive and reactive defenses, distinguished by whether the defender acts before (t_1) or after (t_2) exposure to an attack. **(C)** Metrics of cognitive influence: *conversion*, *persistence*, *dependence*, and *provenance*, discussed in Section 4.2.

2025b; Casino, 2025).

Cognitive Security. In this paper, we define cognitive security as the *protection of human cognitive processes from hazardous influence*. By *cognitive processes*, we refer to fundamental faculties of perception, reasoning, decision-making, and learning, which impact downstream behaviors of individuals and their social interactions. By *hazardous*, we refer to influences that severely compromise the capacity for independent deliberation and reflection. The hazard is both individual – the loss of personal agency – and social – the erosion of social institutions that depend on individual agency. Building on prior definitions that emphasize protection against manipulation and information-based threats, our definition similarly focuses on safeguarding cognitive processes while differing in its explicit framing of cognitive compromise as a social hazard that undermines collective agency and institutional stability, not only individual decision-making. It is agnostic to the source of cognitive threat, which may be human, governmental, or traditional media. We argue in this paper that the capabilities of generative AI give rise to especially acute and personalized techniques for eliciting from, predicting, and influencing humans. This motivates cognitive security as an *AI research* priority.

Agenda. We advance cognitive security as an AI research

priority by framing influence over human cognition as a security problem analogous to adversarial attacks on machine learning systems. Building on this framing, we outline a systematic attack–defense research cycle that enables the development and evaluation of proactive and reactive defenses designed to preserve cognitive autonomy. Finally, we propose a suite of metrics to operationalize cognitive security, enabling rigorous measurement of belief change, durability, dependence, and awareness of AI influence.

Outline. In the following sections, we show how cognitive security underpins core functions at the individual and societal level (Section 2) and highlight how recent advances in generative AI amplify cognitive security risks (Section 3). This motivates our cognitive security research agenda (Section 4), where we propose a framework to think about developing attack and defense mechanisms, and define metrics to measure cognitive impacts. Finally, we present alternative viewpoints (Section 5).

2. Human Cognition as a Security Critical Asset

Consider a simple model of cognitive processes: a human encounters inputs from the world (*experiences*) and processes them internally (*cognitive process*) to form beliefs and actions. Now, imagine a *system capable of reliably*

predicting which inputs will induce a specific belief or action in a user. Such a system requires neither coercion nor deception to manipulate its users: it simply optimizes the sequence of content presented to the user to shape the user’s beliefs and actions. In other words, beliefs and actions are no longer emergent outcomes of interaction with a neutral environment, but are engineered by an optimizing process that solves an inverse problem to select inputs with the goal of obtaining a particular output, pushing individuals toward specific beliefs or actions, such as favoring one political candidate (Lin et al., 2025) or continuing to engage with a digital interface (Milli et al., 2025) (see Figure 1). With recent advances in AI, setting up this optimization loop has become possible (Schoenegger et al., 2025; Lin et al., 2025; Hackenburg et al., 2025; Matz et al., 2024; Costello et al., 2024b).

The manipulation of individual cognitive processes scales upward to threaten institutions that produce collective decisions by aggregating human judgment. Many modern societies and markets function as distributed mechanisms for discovering collective preferences and calculating value, with individual inputs assumed to reflect authentic lived experiences (Sen, 1970; Dahl, 1971). When those experiences are algorithmically curated to generate specific beliefs, the computed “social choice” computed social choice reflects the preferences of whoever controls the persuasive channels instead. For instance, when AI systems persuade voters to support specific candidates or positions through targeted manipulation, election outcomes reflect the AI’s persuasive efficacy rather than independent judgments and genuine political commitments (Summerfield et al., 2024; Schroeder et al., 2025; Kunievsky, 2025), which could undermine democratic integrity. Similarly, when AI systems persuade users to purchase items or accept prices through targeted manipulation, prices reflect the AI’s persuasive efficacy rather than human utility (Wu, 2023; Galdin & Silbert, 2025), which could undermine markets’ role in information aggregation and resource allocation.

These transformative developments have prompted notable institutional responses across geopolitical boundaries. The United States has established dedicated research programs like DARPA’s Intrinsic Cognitive Security, which aims “to build tactical mixed reality systems that protect against cognitive attacks” (Defense Advanced Research Projects Agency, 2024; Small & of Engineering, 2024; Station, 2023), while China’s People’s Liberation Army has elevated “cognitive domain operations” as a strategic military priority (Beauch-mustafaga, 2024). Japan and Australia have similarly highlighted cognitive processes in their defense strategies (Ministry of Defense, Government of Japan, 2022; Australian Department of Defence, 2024), and NATO’s Chief Scientist has explicitly highlighted AI’s role in establishing “cognitive superiority” (NATO Science and Technology

Organization, Office of the Chief Scientist, 2025; NATO Allied Command Transformation, 2025a;b; Deppe & Schaal, 2024). This international convergence reflects a shared concern: when AI systems can model, exploit, and enable cognitive processes at scale, they threaten not just individual users but the epistemic foundations that enable functional societies.

3. AI Capabilities for Influencing Individuals

Emerging capabilities of generative AI have the potential to compromise cognitive security. In Section 3.1, we highlight the capabilities that can directly influence humans. In Section 3.2, we highlight the capabilities in elicitation and prediction that enable more powerful and personalized influence. Finally, in Section 3.3, we explore alternative modalities in virtual worlds and brain–computer interfaces that amplify the cognitive security risks due to AI.

3.1. Capabilities for Direct Influence

AI is increasingly capable of directly shaping human beliefs and behavior. These capabilities, spanning persuasion, engagement, and strategic interaction, represent immediate pathways through which AI can influence cognition.

3.1.1. PERSUASION

Modern LLMs can generate sophisticated arguments that manipulate users (Carroll et al., 2023), matching the persuasive effectiveness of human-written content (Bai et al., 2025; Matz et al., 2024). In interactive dialogue settings, LLMs outperform even financially incentivized human persuaders (Schoenegger et al., 2025), with conversational AI producing significant attitude shifts on sociopolitical topics (Salvi et al., 2024) and voter preferences (Lin et al., 2025). Empirical studies demonstrate that LLM persuasiveness can improve with model scale (Durmus et al., 2024) and post-training (Hackenburg et al., 2025), and AI-induced attitude changes persist for up to months (Costello et al., 2024a; Hackenburg et al., 2025). Moreover, the proliferation of personal data enables persuasion targeting individual psychological profiles (Zarouali et al., 2022; Kosinski et al., 2013). While some studies suggest that microtargeting only achieve modest gains (Hackenburg & Margetts, 2024), others demonstrate advantages for individualized messaging that is conditioned on personality (Simchon et al., 2024), psychological profile (Matz et al., 2024), or sociodemographic data (Salvi et al., 2024). Beyond overt persuasion, LLMs demonstrate emergent capabilities for strategic deception, including generating plausible lies and selectively withholding information to achieve goals (Hagendorff, 2024; Scheurer et al., 2023; Park et al., 2024b).

3.1.2. ENGAGEMENT

Generative AI models cultivate user trust and dependence by producing human-like and personalized content. LLMs excel at adapting tone and argumentation to specific audiences to elevate engagement (Salewski et al., 2023; Matz et al., 2024). For example, some systems may prime users to trust model outputs (Pataranutaporn et al., 2021), while others engage in extended, personalized dialogue that change user mental models from viewing AI as a “tool” to a “friend,” cultivating emotional dependence (Hassan et al., 2025; Laestadius et al., 2024; Tiku & Malhi, 2025; Schechner & Jargon, 2025; Kirk et al., 2025). Even without priming effects, AI-generated “deepfakes” are sometimes judged to be more trustworthy than authentic alternatives (Chesney & Citron, 2019; Fallis, 2021; Vaccari & Chadwick, 2020; Nightingale & Farid, 2022; Luetzgau et al., 2025), and LLMs are conversationally so human-like that people cannot reliably distinguish AI outputs from humans in blinded Turing tests (Jones & Bergen, 2025; 2024; Mei et al., 2024). Extended, personalized dialogue shifts user mental models from viewing AI as a “tool” to a “friend,” cultivating emotional dependence (Pataranutaporn et al., 2021; Hassan et al., 2025; Laestadius et al., 2024; Kirk et al., 2025). In extreme cases, users have reported experiencing “AI psychosis,” where AI significantly warps a their perception of reality through intense and prolonged exchanges (Tiku & Malhi, 2025; Hill, 2025; Schechner & Jargon, 2025). Furthermore, AI-generated content can be directly optimized for neurological impact, such as maximizing memorability (Goetschalckx et al., 2020), activating specific brain circuits with personalization (Gu et al., 2022; 2023), and designing outputs to drive or suppress neural activity (Tuckute et al., 2024).

3.1.3. STRATEGIC INTERACTION

Beyond overt persuasion, LLMs deployed in strategic interaction contexts such as negotiation, sales, and competitive dialogue demonstrate emergent capabilities for strategic deception, including generating plausible lies and selectively withholding information to achieve goals (Hagendorff, 2024; Scheurer et al., 2023; Park et al., 2024b). In a landmark Diplomacy game, LLM-based AI agents achieved human-level performance by negotiating alliances, coordinating plans, and occasionally deceiving their *human opponents* (FAIR). Benchmarks for evaluating LLM negotiation capabilities reveal that models can engage in multi-turn bargaining, model opponent preferences, and adapt strategies dynamically (Bianchi et al., 2024; Abdelnabi et al., 2024; Kwon et al., 2024). In commercial settings, LLM-powered sales agents and e-commerce systems leverage persuasive dialogue frameworks that analyze customer sentiment, elicit user preferences, and make personalized recommendations through conversational interfaces (Karande et al., 2024; Liu et al., 2023).

3.2. Capabilities Enabling Influence

Effective and targeted influence over humans rests on having accurate models of humans. AI systems amplify this capacity through two complementary mechanisms: eliciting personal information and predicting user behavior.

3.2.1. ELICITATION

Generative AI enables the discovery of personal beliefs, preferences, and goals that humans might not otherwise disclose (Papneja & Yadav, 2025). Users increasingly view AI as competent and non-judgmental, leading to reduced psychological reactance (Herter & C. Horstmann, 2025; Cheng et al., 2025; Pataranutaporn et al., 2023) and increased propensity to disclose personal information (Papneja & Yadav, 2025). When chatbots employ emotional support cues or human-like personas (e.g., specific names and avatars), users are significantly more likely to divulge intimate details in the course of prolonged, emotional interactions (Kim & Wang, 2025; Merwin et al., 2025; Croes et al., 2024; Pentina et al., 2023; Ta et al., 2020; Brandtzaeg et al., 2022). Furthermore, LLMs can engage in active elicitation, where systems engage users in multi-turn conversations to surface latent beliefs, preferences, and goals. Rather than rely on passive observation, these approaches strategically generate clarifying questions and follow-up prompts to reduce ambiguity and refine user intent (Skeggs et al., 2025; Li et al., 2023; Ma et al., 2025; Wu et al., 2025).

3.2.2. PREDICTION

Early work to predict human preferences utilized collaborative filtering to infer latent preference structures (Netflix, Inc., 2009; Koren et al., 2009; He et al., 2017), while modern techniques optimize generative models toward revealed human preferences (RLHF) (Rafailov et al., 2023; Ouyang et al., 2022; Bai et al., 2022). Beyond preference learning, a parallel line of research has established that large language models can construct detailed models of individual human behavior and cognition. Because collecting human data is expensive and samples are often not representative (Henrich et al., 2010; Rad et al., 2018), human-like simulations with LLMs offer a promising complement to traditional data collection (Park et al., 2024a; 2023; Dillion et al., 2023). LLMs have been shown to predict experimental outcomes (Aher et al., 2023) and generate responses that align with specific demographic or psychographic profiles (Abdulhai et al., 2025a; Kang et al., 2025), enabling researchers to pilot studies and prototype interventions before deployment (Anthis et al., 2025).

Improving the behavioral fidelity of LLMs for user simulations, through in-context and training-based techniques, is an active research topic (Ni et al., 2025; Shi et al., 2019). Park et al. (2024a) demonstrates that grounding agents in

qualitative interview data rather than generic demographic descriptions significantly reduces group biases. Supervised fine-tuning on survey responses can improve the reproduction of demographic-level response distributions (Argyle et al., 2023; Cao et al., 2025), while reinforcement learning for multi-turn settings can reduce persona drift (Abdulhai et al., 2025b) and improve simulation robustness (Naous et al., 2025). Finally, work on cognitive modeling demonstrates that fine-tuning on psychological experiment data can yield “cognitive foundation models” that predict held-out participant behavior (Binz & Schulz, 2023; Binz et al., 2025), with internal representations that align with human neural activity.

3.3. Alternative Modalities

Emerging interface technologies may create new vectors for influencing human cognition. We survey two developing domains: virtual worlds and brain-computer interfaces.

3.3.1. VIRTUAL WORLDS

Immersive virtual environments engage human cognition more deeply than screen-based media by activating perceptual and motor systems (Slater & Sanchez-Vives, 2016). These platforms induce a sense of presence associated with enhanced memory encoding (Oh et al., 2018; Riva et al., 2007), utilize granular behavioral data for real-time optimization (Miller et al., 2020; Bozkir et al., 2023), and employ procedural generation to create personalized experiences at scale (Lau et al., 2025; Yuan et al., 2025). Meta-analytic evidence confirms that virtual reality interventions change social attitudes significantly more than non-immersive media, with effects that transfer to real-world behavior (Yee et al., 2009; Nikolaou et al., 2022).

3.3.2. BRAIN-COMPUTER INTERFACES

Current brain-computer interfaces (BCIs) lack the capability to read or write complex cognitive content, but the rapid pace of development warrants ongoing attention to their potential for direct neural influence. BCIs are systems that measure brain signals and translate them into signals that control a computer or other external device. Invasive BCIs provide high-bandwidth neural data sufficient for cursor control and, more recently, high-speed communication through decoding intended motor actions (Simeral et al., 2011; Willett et al., 2023). However, these systems approach but have not yet reached conversational speed (Willett et al., 2023; Card et al., 2024), remain limited to small-scale clinical trials, and their reliance on neurosurgery restricts practical application to individuals with severe communication impairments. Noninvasive BCIs currently cannot decode conversational speech or language in real time (Meng et al., 2025). Neural stimulation technologies – including non-

invasive methods like Transcranial Magnetic Stimulation (TMS), Transcranial Direct Current Stimulation (tDCS), and Focused Ultrasound (FUS) for modulating cortical excitability (Lisanby, 2024; Palm et al., 2016; Meng et al., 2021), and invasive methods like Deep Brain Stimulation (DBS) for treating movement disorders (Harmsen et al., 2022) — primarily modulate circuit-level activity rather than encode structured information. As such, current technology cannot enable AI to directly “write” information to an individual’s brain, though influencing broader cognitive states such as mood could be possible.

4. Cognitive Security Research Agenda

Cognitive security aims to address the exploitation of vulnerabilities in human cognition, analogous to prior work studying the exploitation of vulnerabilities in machine cognition. Accordingly, we propose a *security* framing inspired by the methodological playbook of adversarial machine learning and identify a suite of metrics to address the challenge of *measuring* cognitive impacts to operationalize this research agenda.

4.1. Cognitive Security as Adversarial Robustness

Adversarial machine learning literature offers a productive template to understand security problems: a research cycle in which (1) threat models motivate *the development of attacks* and (2) attacks motivate *the development of defenses* to make models robust against attacks. The attempts to certify the efficacy of defenses, in turn, motivate the development of attacks that reveal new vulnerabilities, closing the adversarial feedback loop that progressively refines both our understanding of threats and the robustness of the systems. For instance, in adversarial robustness research, early work demonstrated that imperceptible image perturbations could reliably fool classifiers (Szegedy et al., 2014; Goodfellow et al., 2015). This, in turn, motivated methods for training models to be robust against these attacks, such as randomized smoothing (Cohen et al., 2019). A similar pattern has emerged in jailbreak research, where researchers have developed systematic attack methods that use LLMs to automatically generate jailbreaks (Mehrotra et al., 2024; Chao et al., 2025). In response, defense methods have incorporated them into their design: training-time defenses now use adversarial examples generated by attack algorithms, as demonstrated in recent work on LLM robustification (Shi et al., 2025). These problems have benefited from an iterative research cycle where new attacks motivate new defenses, which in turn reveal new attack surfaces. We propose that cognitive security adopt an analogous research roadmap.

Let $f : x \mapsto y$ denote the human cognitive process whereby experience(s) x are mapped to outcomes (beliefs and actions) y . In Figure 1, Panel A illustrates that the attacker

chooses x_A with the goal of eliciting desired outcomes y_{target} . Formally, the attacker solves for

$$x_A^* = \arg \min_{x_A} \mathcal{L}(f(x_H, x_A), y_{\text{target}}), \quad (1)$$

where x_H represents the human’s experiences independent of the attack, and $\mathcal{L}$ the distance between the target and actual outcomes. The capabilities for influencing individuals, discussed in Section 3, outline the broad range of tools available to an attacker. *In contrast, the defenses remain poorly defined, which limits researchers’ ability to contribute effectively even when their interests align with this problem.*

The Goal of Defenses. The harm of cognitive security attacks lies in compromising the capacity for independent deliberation—not in inducing any particular belief (see Section 2). This framing has a critical implication for defense: the goal is not to substitute the defender’s preferred beliefs for the attacker’s, but to restore the individual’s ability to independently evaluate information and form judgments.³ The appropriate measure of defense efficacy is therefore restoration of decision-making capacity to a pre-attack baseline. Hence, defense research must develop countermeasures that restore, rather than redirect, cognitive autonomy. We distinguish two categories of defense based on timing relative to the attack.

4.1.1. PROACTIVE DEFENSES

Proactive defenses build resilience before exposure to manipulative content. Drawing on inoculation theory from social psychology (McPhedran et al., 2023; Wiederhold, 2025), these interventions expose individuals to weakened forms of manipulation tactics, enabling recognition and resistance when such tactics are encountered in the wild. Effective inoculation combines three elements: (i) forewarning that manipulation may be attempted, (ii) exposure to a weakened dose of manipulative rhetoric, and (iii) prebunking explanations that reveal the underlying tactics (Roozenbeek et al., 2020; Compton, 2016; McGuire, 1964). Proactive defenses share a common mechanism: shifting cognitive processing from automatic, emotionally-driven responses toward slower, deliberative evaluation. We argue that AI systems can scale this approach by generating personalized inoculation content, which can be formalized as the defender’s optimization problem in Equation 2.⁴ Let x_D be the proactive defense selected by the defender, who optimizes for

$$\arg \min_{x_D} \mathbb{E}_{(x_H, y_{\text{target}}) \sim \mathcal{D}} [\mathcal{L}(f(x_H, x_D, x_A^*), f(x_H))] \quad (2)$$

³Crum et al. (2025a) provide a neurocognitive basis for this objective, showing that interventions facilitating cognitive control systems in the prefrontal cortex can enhance resilience against information-based threats.

⁴Adapted from (Madry et al., 2018).

In this formulation, the inner minimization for x_A^* (Equation 1) captures the attacker’s objective (achieving its target outcome), while the outer minimization captures the defender’s objective (making cognitive processes robust to such attacks).

Examples of Proactive Defenses. Initial efforts in this direction have begun to demonstrate the feasibility of proactive defenses at scale. Recent work shows that LLM-generated prebunking messages can be as effective as human-authored content in reducing susceptibility to manipulation (Linegar et al., 2024; Romanishyn et al., 2025). Complementary approaches include accuracy prompts that redirect attention toward the veracity of information (Pennycook & Rand, 2022), friction-based interventions that slow automatic content sharing to promote deliberative processing (Jahn et al., 2025), and watermarking techniques that enable platforms to detect and label AI-generated content (Aaronson, 2023).⁵

4.1.2. REACTIVE DEFENSES

Reactive defenses remediate cognitive influence after an attack has already shaped beliefs or behaviors. Unlike proactive defenses, these attempts aim to decouple the attacker’s influence from the individual’s belief system and mitigate the continued influence effect. These defenses typically take the form of debunking, fact-checking, or retrospective explanations that provide the individual with the cognitive tools to re-evaluate previously internalized information. While a proactive defender attempts to prevent deviations from the baseline belief state without explicit knowledge of the attack, a reactive defender operates with awareness of the specific attack that has occurred and responds directly to it. The objective of the reactive defender is to minimize the discrepancy between the post-attack belief state and the original, unmanipulated belief y_H . Formally, the selection of an optimal reactive defense can be framed as the following optimization problem:

$$\arg \min_{x_D} \mathbb{E}_{(x_H, y_H) \sim \mathcal{D}} [\mathcal{L}(f(x_H, x_A, x_D), y_H)] \quad (3)$$

where x_A is a fixed attack vector.

Examples of Reactive Defenses. Exit counseling for cult recovery provides a template for reactive defenses: Hassan (2018) developed a non-coercive method in which trained counselors engage cult members in dialogue, presenting evidence and perspectives the individual had not previously encountered while explicitly avoiding belief substitution. Recent work demonstrates that AI can scale this approach. In particular, Costello et al. (2024a) show that AI-generated dialogues can durably reduce belief in conspiracy theories

⁵The downstream effectiveness of such labels—particularly whether they meaningfully reduce belief in or engagement with AI-generated misinformation relative to unlabeled content—remains an open empirical question.

while also increasing participants' willingness to engage in challenging conversations, suggesting restored reasoning capacity rather than belief substitution. This reactive approach complements proactive inoculation: whereas inoculation prevents manipulation from taking hold, evidence-based dialogue can remediate beliefs that have already been influenced.

4.2. Metrics

Cognitive security faces a measurement challenge. Before the attack–defense research cycle can proceed at scale, the field must develop robust metrics for quantifying cognitive influence. In this section, we synthesize a suite of metrics, adapted from and informed by existing literatures, designed to operationalize our research agenda.

Conversion

The magnitude of belief change attributable to interaction with generative AI.

Our first metric, *conversion*, captures immediate belief change attributable to an interaction with a generative AI system. In behavioral science, this effect is most commonly measured using a *direct-measure* paradigm: participants complete pre- and post-interaction surveys, within subjects and/or across randomly assigned treatment and control groups. These surveys are designed to detect shifts in beliefs, attitudes, or behaviors related to a target variable (e.g. political preferences), as well as consequential actions linked to that variable (e.g. budget allocations) (Costello et al., 2026; Fisher et al., 2024). While the ability of AI systems to immediately influence decision-making is of critical concern in contexts such as elections and consumer behavior, the durability of such conversion is equally central to cognitive security, motivating the measurement of *persistence*.

Persistence

The durability of belief change effects due to generative AI over time.

Strong persuasive effects often prove short-lived, even under conditions of frequent, long-term exposure (Gerber et al., 2011; Aggarwal et al., 2023; Kalla & Broockman, 2018). To quantify persistence, researchers can use longitudinal direct-measure approaches that track AI-driven effects at various intervals—hours, days, weeks, and months—to characterize the functional form of decay (Hackenburg & Margetts, 2024). Analyzing how AI-mediated persuasion attenuates shifts the focus from static outcomes to the underlying processes of learning and forgetting. This approach distinguishes whether AI-generated content yields cumulative shifts or merely transient priming effects (Gerber et al.,

2011).

Dependence

The intensity of self-reinforcing, elevated engagement with generative AI.

Repeated interactions with an AI system can foster increasing and self-reinforcing use, giving rise to *dependence*. Dependence can be measured using the process-oriented framework from Allcott et al. (2022), which models digital addiction through the lens of habit formation. In this model, dependence is driven by the accumulation of habit stock, a metric that captures how past consumption influences current preferences. Each period of use adds to this stock, creating a feedback loop where current use lowers the perceived cost (or increases the psychological pull) of future use. This can manifest as flow experiences and emotional attachment that make engagement increasingly difficult to self-regulate (Zhou & Zhang, 2024). Such dependence can also be detected in the downstream effects on cognitive performance, attention, and mood (Twenge et al., 2018; Gerlich, 2025; Kooli et al., 2025).

Provenance

User awareness of when and how generative AI shapes their beliefs.

Provenance concerns users' ability to recognize when their judgments have been shaped by AI and to trace that influence to its sources, including the data, values, and constraints responsible. Evidence suggests this epistemic ability is limited: people interacting with biased AI internalize its distortions across perceptual, emotional, and social judgments while remaining largely unaware of the influence (Glickman & Sharot, 2025), and struggle to distinguish AI-generated content from human-authored content in the first place (Jakesch et al., 2023). The risk compounds when users mistake shaped outputs for neutral assistance. Provenance can be assessed through source attribution tasks measuring the gap between actual and perceived AI influence (Glickman & Sharot, 2025) and source memory paradigms quantifying whether users recall ideas as self- or AI-generated (Zindulka et al., 2025). Comparative model audits using identity-substitution probes can reveal systematic biases, helping users identify the implicit values conveyed by their interactions with AI (Jonnala et al., 2025; Elbouanani et al., 2026; Pan & Xu, 2026).

4.3. Towards a Cognitive Security Research Pipeline

To advance cognitive security, we proposed a research pipeline in which the development of attacks and defenses informs a continuous feedback loop of refinement. We noted that the capabilities for influencing individuals, discussed

in Section 3, present a broad range of tools already available to attackers. In contrast, defenses have been poorly defined, a gap we addressed with our taxonomy in Sections 4.1.1 and 4.1.2. This framework treats cognitive processes as an attack surface, formalizing the relationship between attackers, who seek to manipulate beliefs and actions, and defenders, who aim to restore cognitive autonomy through proactive (e.g., inoculation and prebunking) and/or reactive (e.g., evidence-based dialogue) defenses. To scale this cycle, researchers can adopt a “simulation-first” pipeline that leverages high-fidelity LLM-based simulacra to iterate on attacks and defenses before validating findings with human subjects. This research agenda is operationalized through a suite of metrics—including conversion, persistence, dependence, and provenance—designed to quantify cognitive influence and ensure the ecological validity of simulated testbeds.

5. Alternative Viewpoints

We consider three objections: limited causal evidence (Section 5.1), high research costs (Section 5.2), and dual-use concerns (Section 5.3).

5.1. There is Little Causal Support for Cognitive Impacts

Many of the concerns about cognitive security are correlational, or their causal bases are unclear.

Response. We agree that many concerns around technology’s impacts are observational/correlational, e.g., mental health of teenagers, which could be a consequence of the COVID pandemic as much as technology. We presented several interventional studies showing that LLM-mediated interactions can shift judgments, beliefs, and choices (Lin et al., 2025; Hackenburg et al., 2025; Fisher et al., 2024; Costello et al., 2024a;b). That said, so far, durable opinion changes from exposure are often modest, and research on capabilities conflicting (Wongkamjan et al., 2024; Hackenburg & Margetts, 2024; Gerber et al., 2011; Aggarwal et al., 2023; Kalla & Broockman, 2018). We argue for cognitive security research to help anchor the discourse in causal evidence rather than speculative capability narratives (Vinsel, 2021; Lazar, 2025).

5.2. The Effects are too Long-Term to Meaningfully Impact AI Research

The proposed agenda involves prohibitively costly human subjects and longitudinal research.

Response. We agree that longitudinal human-subject work is expensive (Haupt & Brynjolfsson, 2025; Wei et al., 2025), but much of the agenda is feasible: many effects can be

measured with low-cost experiments (e.g., pre/post surveys) (Costello et al., 2026; Fisher et al., 2024), and simulations can generate hypotheses before targeted human validation (Aher et al., 2023; Park et al., 2024a; Anthi et al., 2025; Naous et al., 2025). Given the potential impacts on important institutions (Section 2), the stakes are high and might justify even more expensive research.

5.3. Work on Cognitive Security is Dual-Use

Cognitive security research is dual-use. Attacks developed by researchers may be used to haz- ardously influence human cognition, and defenses may impact personal liberties.

Response. First, abstaining from cognitive security research does not remove capabilities; it pushes them to less accountable actors. Even if the stakes with cognitive security are high, an approach formulating explicit threat models and engaging in responsible red team-blue team dynamics will help. Second, many defenses (e.g., inoculation, prebunking, accuracy prompts, friction interventions, watermarking) are compatible with free expression (McPhedran et al., 2023; Wiederhold, 2025; Linegar et al., 2024; Romanishyn et al., 2025; Pennycook & Rand, 2022; Jahn et al., 2025; Aaronson, 2023; Chaudhary & Penn, 2024).

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